

Reward-Free Self-Prediction Drops Exogenous Control-Relevant Features: A Controlled Study

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Abstract

Joint-embedding predictive (JEPA-style) objectives learn representations by predicting future latents, and in doing so they drop features that are *exogenous*—uncontrollable by the agent—yet control-relevant, even when those features are trivially encodable, because the objective prices temporal predictability rather than control-relevance. We isolate this failure in a controlled 2×2 experimental design that varies feature controllability and relevance independently, equipped with a *predictability knob* that decouples a feature’s temporal predictability from its control-relevance. Across six objectives—reconstruction, JEPA, action-conditioned JEPA, controllability-based JEPA, inverse dynamics under a random policy, and reward-grounded JEPA—every reward-free self-prediction objective fails to retain the *exogenous control-relevant feature* (linear-probe accuracy ≈ 0.51 , InfoNCE mutual information ≈ 0 nats), whereas reward-grounded prediction retains it selectively (probe 1.00, MI = 0.693 nats = $\log 2$, the maximum for a one-bit feature). The fix is label-efficient and robust: as little as 2% of transitions carrying a reward label suffices to recover the feature, the effect holds across two environments with different surface forms, and it persists across latent dimensions from 16 to 1024. At JEPA convergence the empirical bisimulation error between analytically distinct feature classes is 1.893 out of a maximum of ≈ 2.0 —a near-maximal violation—indicating that the dropped information is exactly the reward-relevant structure that bisimulation theory says an optimal representation should keep.

1 Introduction

A representation can fail a downstream task not because it lost something irrelevant, but because it lost something it could not predict. Consider a feature that is simultaneously *control-decisive* (the optimal action depends on it), *trivially encodable* (it appears as a salient, localized patch in the observation), and yet *temporally unpredictable* (its value next step is close to a coin flip). A self-prediction objective, which is rewarded for predicting its own future latents, receives no gradient that asks it to carry such a feature: the feature is noise from the objective’s point of view, even though it determines the agent’s payoff. The feature we study is not irrelevant—it is control-decisive—and not uncontrollable; it is merely unpredictable. Yet this is sufficient for a JEPA-style objective to discard it.

Joint-embedding predictive architectures predict future latent states rather than reconstructing pixels (Assran et al., 2023, 2025). Discarding “unpredictable detail” is intentional and is precisely what makes these objectives scalable and robust to nuisance texture. The question this paper asks is narrow but consequential: does “unpredictable” always coincide with “irrelevant”? When the two come apart, a representation optimized for predictability will throw away exactly the information a controller needs.

This tension is sharp against bisimulation theory. Zhang et al. (2020) argue that an optimal control representation should keep features that distinguish states by their reward-relevant futures

and collapse everything else; a JEPA objective instead keeps what is predictable. For a feature that is reward-relevant but temporally unpredictable, these two criteria point in opposite directions: bisimulation says keep it, self-prediction says drop it. The disagreement is not hypothetical—it is realized by a single cell of a simple taxonomy.

We organize features along two binary axes—*controllability* (does the agent’s action move the feature?) and *relevance* (does the feature set the reward or the optimal action?)—and study the resulting four cells. (i) Controllable and relevant: every objective keeps it (trivial). (ii) Controllable and irrelevant: controllability-based methods waste capacity on it while bisimulation drops it. (iii) Uncontrollable and irrelevant: the classic exogenous distractor, studied extensively. (iv) **Uncontrollable and relevant**: the *exogenous control-relevant* feature, where the failure lives. Prior work has studied cells (i)–(iii) extensively; cell 4 has been named as a risk but, to our knowledge, not measured in controlled isolation.

Contributions.

- **A controlled isolation of the exogenous control-relevant case.** We provide the first controlled isolation of cell 4 (exogenous + relevant) using a predictability knob that independently manipulates a feature’s temporal predictability and its control-relevance, ruling out “the objective dropped it because it was irrelevant” as an explanation.
- **Empirical measurement of the failure and a selective fix.** We find that all four reward-free self-prediction objectives—JEPA and its action-conditioned, controllability-based, and inverse-dynamics variants—fail on cell 4 (linear-probe accuracy ≈ 0.51 , mutual information ≈ 0 nats, effective rank 38–42 confirming the representation has *not* collapsed), while reward-grounded prediction fixes it selectively (probe 1.00, MI = 0.693 nats, retaining exactly the relevant cells 1 and 4). Reconstruction retains it too, but only by indiscriminately keeping every feature, including the irrelevant distractors.
- **Practical scaling findings.** The fix needs as little as 2% of transitions reward-labeled; the failure persists from 16 to 1024 latent dimensions; and it replicates across two environments with different surface forms.

Figure 1 (Section 4.1) summarizes the central result across both environments: every reward-free self-prediction objective sits at chance on the exogenous control-relevant feature, and only reward grounding recovers it among the practical methods.

2 Background

2.1 JEPA and latent self-prediction

A joint-embedding predictive architecture trains an encoder to predict the latent of a future (or masked) observation from the latent of the present one, without reconstructing pixels (Assran et al., 2023). Because the loss only rewards predicting the part of the future that *is* predictable, a feature whose future value is unpredictable receives no signal to be retained. This is a deliberate design choice—it is what lets these models ignore nuisance detail—and it is central to the program of latent-space world models for autonomous agents (LeCun, 2022).

2.2 Bisimulation metrics

Bisimulation metrics define a distance on states that respects reward and transition structure: two states with the same reward-relevant future are at distance zero, and states differing in a reward-relevant feature are at positive distance (Ferns et al., 2004). Zhang et al. (2020) (DBC) showed that representations trained to match a bisimulation metric are substantially more robust to task-irrelevant distractors than reconstruction- or contrastive-based representations, because the metric explicitly factors out reward-irrelevant variation. The property we rely on is the converse: a reward-relevant feature has *positive* bisimulation distance and must be kept. Our Observation 1 (Section 5) measures the gap between the analytical bisimulation distance of the exogenous control-relevant feature and the class separation a trained JEPA latent actually realizes.

2.3 A feature taxonomy and the exogenous control-relevant cell

We treat the two axes as definitions. A feature is *controllable* if c_{t+1} is a function of the action a_t (the agent can move it) and *uncontrollable/exogenous* if c_{t+1} evolves independently of a_t . A feature is *relevant* if it determines the reward, and hence the optimal action a^* , and *irrelevant* if it has no effect on reward. The four cells of the resulting taxonomy are not a new framework but a controlled experimental design that lets us vary one property at a time. Our focus is cell 4, the uncontrollable-but-relevant case. It is the hard case for a specific reason: controllability-based methods drop it because the agent cannot move it; prediction-based methods drop it because the agent cannot predict it; only reward-based methods have a signal that distinguishes it from an irrelevant exogenous feature.

3 Setup

3.1 Environments

The primary environment, `QuadrantEnv`, emits 12×12 grayscale observations. A single 3×3 corner patch encodes the target feature bit, superimposed on a rich, deterministically-advancing multi-frequency background (four spatial frequencies with a deterministic phase progression). The background is therefore *predictable* structure that a self-prediction encoder can and does model, which is what makes the contrast with the unpredictable feature patch clean.

We replicate the central result on two surface forms. `SwitchColorEnv` (32×32 observations) encodes the bit in patch intensity over a sinusoidal grating background; `GridWorldHiddenRuleEnv` places the patch in the bottom-right corner over a drifting texture with an agent marker. The two surface forms share the cell-4 logic but differ in resolution, texture statistics, and intrinsic dimensionality, which lets us separate the objective-level effect from any one rendering.

Feature dynamics follow a single specification. Controllable features toggle on the action taken; exogenous features evolve with $P(c_{t+1} = c_t) = p_{\text{repeat}}$, the predictability knob (default $p_{\text{repeat}} = 0.5$, i.e. maximally unpredictable). The optimal action is $a^* = c_t$ and the reward is 1 iff the action matches the relevant feature bit. A helper `make_cell(cell, predictability)` instantiates the single-feature specification for each quadrant cell. Unless otherwise stated, the main experiments train for 4000 steps with batch size 128, learning rate 10^{-3} , latent dimension 128, and three seeds $\{0, 1, 2\}$ (Appendix A).

The five scientific invariants. The experimental protocol is built around five invariants, stated here because they are the conditions under which the comparison across objectives is meaningful.

1. **Byte-identical encoder.** The encoder architecture is identical (matching `state_dict` keys and shapes) across every objective, so any difference in retention is attributable to the objective, not the encoder. This is asserted automatically (Appendix D).
2. **Anti-collapse regularization.** JEPA objectives use VICReg-style variance (and covariance) regularization (Bardes et al., 2022); we empirically confirm the latent has not collapsed by reporting effective rank alongside every probe (Section 3.4).
3. **BYOL-style target.** The prediction target is produced by an exponential moving-average (EMA) copy of the online encoder (Grill et al., 2020) (decay 0.996) with a normalized prediction loss.
4. **Honest probing.** All probes and regret estimates read the *online* encoder (never the EMA target) on a disjoint evaluation seed; this invariant is asserted automatically.
5. **Configurable capacity and idempotent caching.** The latent dimension is configurable (here 128, swept 16–1024 in the capacity study) and “skip-if-done” is keyed on result JSONs rather than checkpoints.

3.2 The predictability knob

The knob $p_{\text{repeat}} \in [0.5, 1.0]$ controls only predictability. At $p_{\text{repeat}} = 0.5$ the feature is an i.i.d. Bernoulli draw (maximally unpredictable); at $p_{\text{repeat}} = 1.0$ it is constant (perfectly predictable). Crucially, control-relevance is held fixed across the whole range: the feature always determines the reward. This decoupling is the methodological core of the study—it lets us ask whether JEPA’s information loss is a function of *predictability* rather than *relevance*, and rules out “the objective merely dropped an irrelevant feature” as an explanation. All main results are reported at $p_{\text{repeat}} = 0.5$. As a sanity check, sweeping the knob in `SwitchColorEnv` shows JEPA recovers the feature only as it becomes predictable, while reconstruction keeps it throughout (Appendix A).

3.3 Objectives

We compare six objectives plus an oracle upper bound. The encoder is byte-identical across all of them (invariant 1); they differ only in which signal the objective consumes (Table 1).

For `jepa_invdyn` the policy matters. Under the random-policy stream used in the failure experiments, the actions carry no information about an exogenous feature, so the inverse-dynamics head has nothing to anchor cell 4 to and drops it. Under an informative policy whose actions correlate with the feature (the condition shown in the full matrix, Table 3 in Appendix C), the same objective can retain it—but that is oracle-grade action supervision, not unsupervised rescue. We therefore report `jepa_invdyn` as a reward-free objective evaluated under random actions, and flag the informative-action result separately.

3.4 Metrics

We report four metrics, each chosen to answer a distinct question.

- **Linear probe accuracy** (primary): can the feature c be decoded by a linear classifier on the frozen latent z ? Chance is 0.50 and we use a retain threshold of 0.75.
- **InfoNCE mutual information:** a lower-bound estimate of $I(Z; c)$ in nats trained on a held-out split, with theoretical maximum $\log 2 \approx 0.693$ for a one-bit feature (van den Oord et al., 2018). It corroborates the probe without sharing its decision boundary.

Table 1: The six objectives plus the oracle upper bound. The encoder is byte-identical across all; each differs only in the signal its objective consumes. `jepa_reward` is the proposed fix; `oracle` is an upper bound that consumes ground-truth labels and is not a practical method.

Objective	Signal consumed	Key property
<code>recon</code>	pixels	reconstruction upper bound; keeps everything
<code>jepa</code>	none (latent self-prediction)	the baseline under study
<code>jepa_ac</code>	actions taken (predictor on a_t)	action-conditioned; still reward-free
<code>jepa_ctrl</code>	actions taken (inverse-dynamics head)	controllability-based; reward-free
<code>jepa_invdyn</code>	actions taken (random policy)	informative only if the policy correlates with c
<code>jepa_reward</code>	reward labels (+ optional bisim term)	the proposed fix
<code>oracle</code>	ground-truth feature labels	upper bound; not a practical method

- **Control regret:** oracle reward minus the reward a policy achieves, on the `SwitchColor` predictability sweep. We use the *independent* rollout variant—a separate behaviour-cloned MLP policy rolled out in the environment, distinct from the linear probe—so that the regret-versus-information comparison (Section 5) is non-circular.
- **Effective rank:** exp of the entropy of the normalized singular values of the latent matrix, reported alongside every probe so that we can distinguish *selective dropping* (high rank, the feature is gone but the representation is rich) from *representational collapse* (low rank, everything is gone). We only describe a result as selective dropping when effective rank exceeds 5.

4 Results

4.1 All reward-free self-prediction objectives drop the exogenous control-relevant feature

Every self-prediction objective that does not consume a reward signal fails to retain the exogenous control-relevant feature, in both environments, across three seeds. The complete picture is in Figure 1 and Table 2. On `QuadrantEnv`, `jepa` reaches linear-probe accuracy 0.51 ± 0.02 with mutual information 0.00 ± 0.00 nats; `jepa_ac` 0.52 ± 0.03 ; `jepa_ctrl` 0.52 ± 0.02 ; and `jepa_invdyn` (random policy) 0.52 ± 0.02 , all at chance with essentially zero retained information (Table 2). In contrast, `jepa_reward` reaches 1.00 ± 0.00 with 0.69 ± 0.00 nats—the one-bit ceiling—matching the `recon` and `oracle` upper bounds.

The picture is the same on `GridWorldHiddenRuleEnv`, with two informative wrinkles. First, the order is preserved: `jepa` sits at 0.51 ± 0.02 and `jepa_reward` at 1.00 ± 0.00 . Second, the action-using objectives are noticeably noisier on this richer surface form—`jepa_ac` reaches 0.61 ± 0.08 and `jepa_invdyn` 0.60 ± 0.10 (Table 2). The large standard deviations are themselves the point: on some seeds the random actions happened to carry marginal information about the rule bit, which is exactly what the `jepa_invdyn` caveat predicts—its retention tracks action informativeness rather than the feature itself. We therefore report these as mean $\pm$ std and never as a mean alone.

Observation 1 (empirical): `exo+rel` feature dropped by pure prediction, rescued by reward grounding
 Cell 4 (uncontrollable + reward-relevant), $p=0.5$, 3 seeds, random-action policy

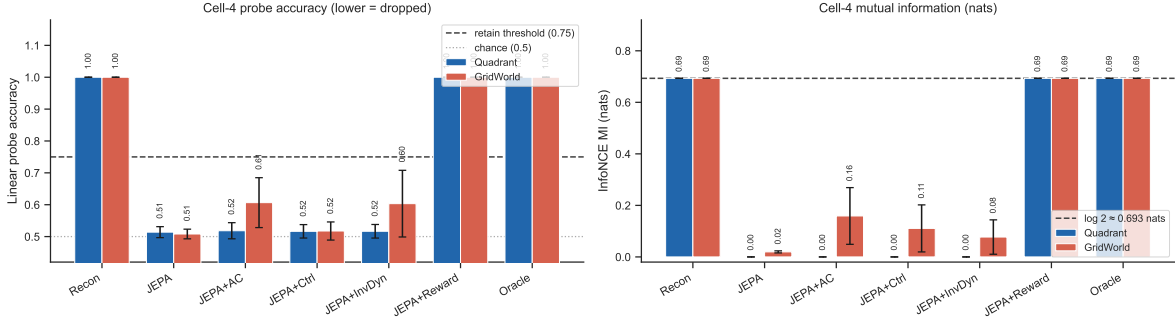


Figure 1: **Cell-4 failure across both environments.** Left: linear-probe accuracy (chance = 0.50, retain threshold = 0.75) for each objective on the exogenous control-relevant feature. Right: InfoNCE mutual information (nats; $\log 2 \approx 0.693$ is the maximum for a one-bit feature). Every reward-free self-prediction objective—including action-conditioned JEPA, controllability-based JEPA, and inverse dynamics with random actions—fails to retain the feature (probe at chance, $MI \approx 0$). Reward-grounded prediction recovers it at the theoretical ceiling, matching the `recon` and `oracle` upper bounds. Bars are mean $\pm$ std over three seeds.

The feature is dropped, not collapsed. On `QuadrantEnv` every dropping objective retains a high effective rank, between 38 and 42; the representation is rich, it has simply discarded this one feature. On `GridWorldHiddenRuleEnv` the dropping objectives have effective rank near 3 (range 2.75–4.61 across all cell-4 runs), but this reflects the intrinsic dimensionality of the simpler images rather than collapse: `recon` retains the feature perfectly at effective rank 3.30. “Low-rank environment” and “collapsed representation” are distinct, and only the latter would undercut the claim; here the environment is simply low-dimensional while the feature-dropping is selective.

The full objective $\times$ cell matrix (Figure 2, with numerical values in Table 3, Appendix C) places cell 4 in context: across the seven rows, the only objectives that retain the exogenous control-relevant feature are `jepa_reward` and the two upper bounds (`recon`, `oracle`).

4.2 Reward grounding is selectively relevant, not selectively controllable

The fix does not merely retain cell 4; it retains the right *axis*. `jepa_reward` keeps both relevant cells—cell 1 (controllable) and cell 4 (exogenous)—and drops both irrelevant cells—cell 2 and cell 3 (Figure 2, Table 3). Its mutual information is 0.693 nats exactly where the probe is 1.00 and 0 nats exactly where the probe is ≈ 0.51 , with zero variance across seeds. Its selectivity axis is relevance.

Contrast this with the controllability-based objective. `jepa_ctrl` keeps cells 1 and 2 (both *controllable*, relevant or not, at ≈ 0.67) and drops cells 3 and 4 (both exogenous). Its selectivity axis is controllability—the wrong axis for control performance, because it spends capacity on a controllable-irrelevant feature while discarding the exogenous-relevant one. The distinction between controllability-based and reward-based selectivity is not semantic: `jepa_ctrl` fails to retain the exogenous control-relevant feature while wasting capacity on the controllable-irrelevant one. (We note that the `jepa_ctrl` cell-1/2 value of 0.67 ± 0.23 is driven by a single seed reaching 1.00 while the others stay near chance, so it is a partial and unstable recovery, not a clean retain; Table 3.)

Table 2: Cell-4 failure across both environments (mean $\pm$ std, 3 seeds, random-action policy). Probe acc < 0.75 = feature dropped. MI in nats; $\log 2 \approx 0.693$ = fully retained. Note large std for `jepa_invdyn`: its rescue tracks action informativeness rather than the feature itself.

Objective	Quadrant		GridWorld	
	Probe acc	MI (nats)	Probe acc	MI (nats)
<code>recon</code>	1.00 ± 0.00	0.69 ± 0.00	1.00 ± 0.00	0.69 ± 0.00
<code>jepa</code>	0.51 ± 0.02	0.00 ± 0.00	0.51 ± 0.02	0.02 ± 0.00
<code>jepa_ac</code>	0.52 ± 0.03	0.00 ± 0.00	0.61 ± 0.08	0.16 ± 0.11
<code>jepa_ctrl</code>	0.52 ± 0.02	0.00 ± 0.00	0.52 ± 0.03	0.11 ± 0.09
<code>jepa_invdyn</code>	0.52 ± 0.02	0.00 ± 0.00	0.60 ± 0.10	0.08 ± 0.07
<code>jepa_reward</code>	1.00 ± 0.00	0.69 ± 0.00	1.00 ± 0.00	0.69 ± 0.00
<code>oracle</code>	1.00 ± 0.00	0.69 ± 0.00	1.00 ± 0.00	0.69 ± 0.00

4.3 The failure is not a capacity problem

Increasing latent capacity does not rescue the feature for JEPA. As the latent dimension grows from 16 to 1024, JEPA’s cell-4 linear-probe accuracy moves only from 0.510 to 0.558 on `QuadrantEnv`—never crossing the 0.75 retain threshold (Figure 3, Table 4). The slight upward drift at very large dimensions is small and plausibly reflects marginal saturation of the covariance regularizer rather than genuine retention. `jepa_reward` holds at 1.00 from dimension 16 through 512 and dips to 0.907 (`Quadrant`) / 0.932 (`SwitchColor`) only at 1024, a training-budget artifact at the largest scale rather than a capacity floor. The cell-4 drop is a property of the objective, not of the encoder’s size.

4.4 Reward grounding is label-efficient

The reward signal is also cheap. The smallest reward-label fraction for which `jepa_reward` crosses the retain threshold is 0.02—one labeled transition in fifty, at mean accuracy 0.77 (Figure 4, Table 4). At a fraction of 0.01 (one in a hundred) the probe is already 0.72, substantially above the chance level of 0.50. This efficiency suggests that the sparse reward signals available in many real environments may be sufficient to prevent the failure even when dense reward annotation is not available.

5 Observation 1 (Empirical)

Observation 1 (empirical). At convergence of the JEPA self-prediction objective on the exogenous control-relevant cell (cell 4, $p_{\text{repeat}} = 0.5$), the class-conditional centroid distance between $c = 0$ and $c = 1$ state representations in the learned latent space is 0.105, while the analytically-derived on-policy bisimulation distance between the same classes is 1.0. The *bisimulation error*—the gap between the analytical distance and the realized latent distance—is 1.893 out of a maximum of ≈ 2.0 . The oracle and reconstruction baselines realize 1.998 of the maximum class separation; JEPA realizes about 5% of the oracle’s.

The cell-4 feature has analytical on-policy bisimulation distance $1.0 > 0$: the immediate-reward gap is 1.0 and the discounted exogenous-transition term vanishes at $p_{\text{repeat}} = 0.5$, so bisimulation theory says the feature must be kept. Only the reward-grounded oracle and the pixel-grounded

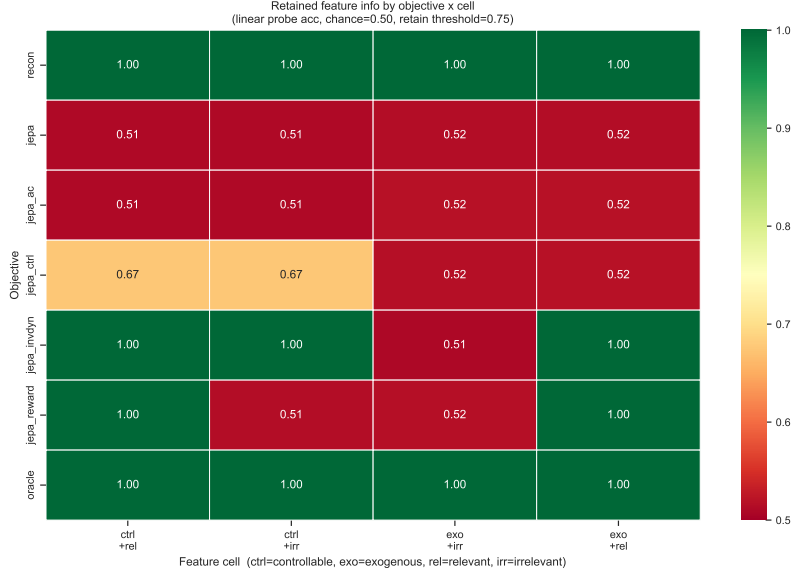


Figure 2: **Full quadrant matrix** of linear-probe accuracy per objective $\times$ cell on **QuadrantEnv**. Green denotes retained (≥ 0.75), red denotes dropped (< 0.75). **jepa_reward** retains exactly the relevant cells (1 and 4) and drops the irrelevant ones (2 and 3); **jepa_ctrl** partially recovers the controllable cells (1–2, ≈ 0.67) and drops the exogenous ones; all purely self-predictive objectives drop every feature under random-policy data. Cell 4 is the column where only reward grounding survives.

reconstruction baseline realize that separation in latent space (class distance ≈ 1.998); the JEPA latent realizes only 0.105, about 5% of the oracle’s, and its linear-probe accuracy sits at chance (0.49). The bisimulation error—oracle minus JEPA, 1.893—stays near-maximal at convergence (Figure 5). Latent self-prediction has no gradient toward an unpredictable-but-relevant feature and collapses it even though it is trivially encodable. These proposition numbers are a single-seed point estimate under full training and are reported as such.

Zhang et al. (2020) relate bisimulation-aligned representations to lower control regret, and Figure 6 confirms that link with two measurements that are both independent of the linear probe: an InfoNCE estimate of $I(Z; c)$, and the regret of a separate behaviour-cloned policy rolled out in the environment. Across the predictability sweep the two are almost perfectly anti-correlated (Pearson $r = -0.999$): every run that retains the feature ($MI \approx \log 2$) achieves zero control regret, while every run that drops it ($MI \approx 0$) is stuck at chance-level control (≈ 0.49), with no run in between. Because neither axis is derived from the probe, this is independent evidence—not the definitional $\text{regret} = 1 - \text{accuracy}$ identity—that the information a reward-free self-prediction objective discards is exactly the information a controller needs. The relationship is a sharp threshold rather than a smooth dose–response, as expected for a one-bit feature. Together with Observation 1—the near-zero class separation and near-maximal bisimulation error measured directly from the latent geometry—this shows the failure is structural rather than incidental.

We emphasize that Observation 1 is an empirical measurement in a controlled synthetic environment, not a formal proof; the analytical bisimulation distance is a closed-form derivation, but the claim that JEPA collapses the feature is measured from trained models on a finite budget, and its generality to large pretrained JEPA models is an open question.

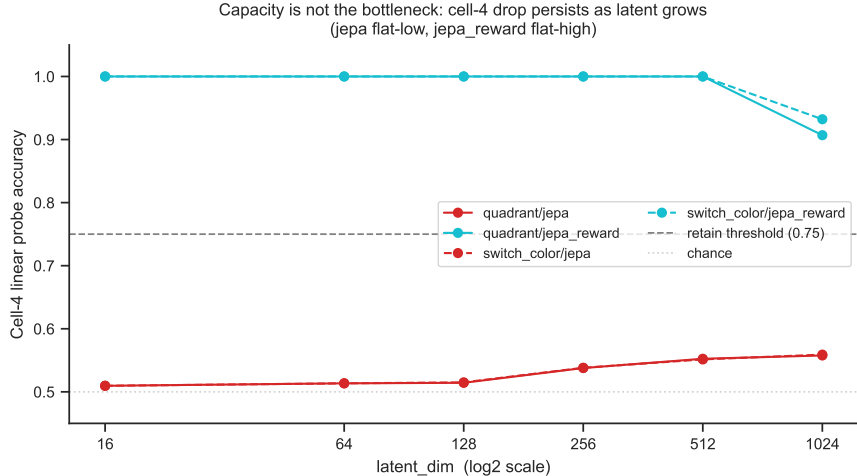


Figure 3: **Cell-4 retention versus latent dimension** for `jepa` and `jepa_reward` on `QuadrantEnv` and `SwitchColor`. JEPA stays near chance across every latent dimension from 16 to 1024, never approaching the retain threshold. `jepa_reward` stays near perfect from 16 to 512, with a slight dip at 1024 attributable to training budget at that scale. The failure is objective-structural, not architectural.

6 Related Work and Discussion

6.1 JEPA and world models

I-JEPA (Assran et al., 2023), V-JEPA 2 and V-JEPA 2-AC (Assran et al., 2025), and DINO-WM (Zhou et al., 2024) establish the latent-prediction objective and demonstrate its empirical benefits for representation learning and planning on pre-trained visual features. These works motivate the objective; our work is complementary, asking what information is structurally at risk under it.

6.2 Bisimulation-based representation learning

DBC (Zhang et al., 2020) established that bisimulation-aligned representations are far more robust to task-irrelevant distractors than reconstruction, by explicitly factoring out reward-irrelevant variation. DeepMDP (Gelada et al., 2019) learns latent models with reward and transition losses that provably bound value error, and action-bisimulation (Rudolph et al., 2024) extends bisimulation to action-conditioned settings. These lines established that the reward signal shapes which structure a representation keeps; our contribution is the controlled isolation of the specific cell where self-prediction and bisimulation conflict—exogenous and relevant—and a quantitative measurement of the information gap.

6.3 Closest prior work

Bisimulation on JEPA world models (arXiv:2602.18639). Toso et al. (2026) apply a bisimulation objective to JEPA world models to handle distractors. We differ in the cell we target: their distractors are exogenous and *irrelevant* (cell 3), the classic distractor the agent neither controls nor needs, whereas we target the exogenous and *relevant* cell 4. Their bisimulation grounding and our reward grounding are related fixes, but they are motivated by different failure modes.

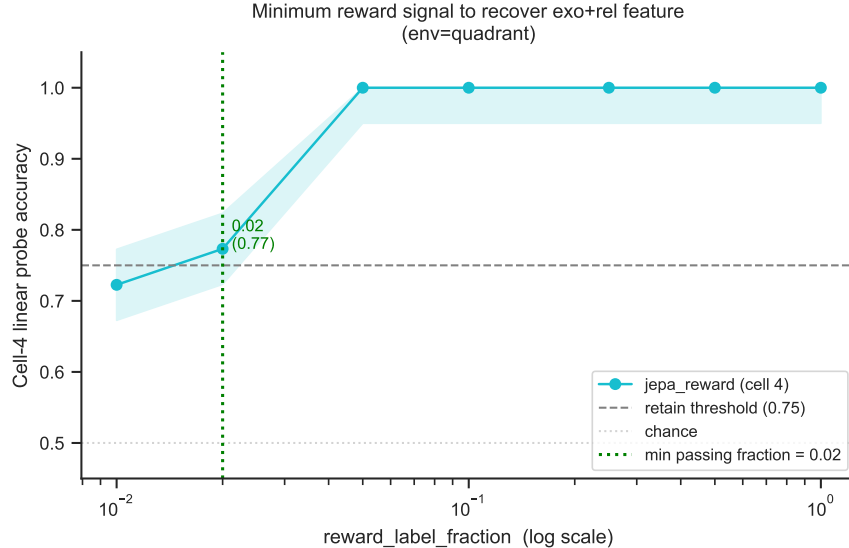


Figure 4: **Cell-4 retention versus the fraction of reward-labeled transitions** (log scale, `QuadrantEnv`). The retain threshold (0.75) is first crossed at a reward-label fraction of 0.02—two percent of transitions (mean accuracy 0.77). Even 1% of labeled transitions yields linear-probe accuracy 0.72, already well above the chance level of 0.50. The reward signal is highly efficient for recovering the exogenous control-relevant feature.

Sensorimotor world models (arXiv:2606.20104). Ivashkov et al. (2026) add an inverse-dynamics regularizer to retain action-relevant features. Our `jepa_invdyn` (random-policy) experiment shows directly that when actions do not encode the exogenous-relevant feature, inverse dynamics fails: cell-4 retention is only 0.52 ± 0.02 on `QuadrantEnv` (Table 2). That fix works for controllable features but does not generalize to exogenous-relevant ones without an informative policy.

A critique of world models (arXiv:2507.05169). Xing et al. (2025) name task-relevant information loss as a concern for JEPA-style models. We make that concern concrete by measuring it—mutual information, bisimulation error, and probe accuracy—in controlled isolation, and by exhibiting a selective fix.

6.4 Limitations

- All results are in controlled synthetic environments with 12×12 (and 32×32) observations. Real-model transfer to large pretrained JEPA encoders (e.g. V-JEPA 2-AC) has not been tested and is future work.
- Observation 1 is empirical, not a proved theorem; a theorem would require explicit assumptions about the learning dynamics and the environment class.
- The gridworld environment has intrinsically low-rank observations (effective rank ≈ 3); the effective-rank anti-collapse check is calibrated for the quadrant environment and should be interpreted differently there.

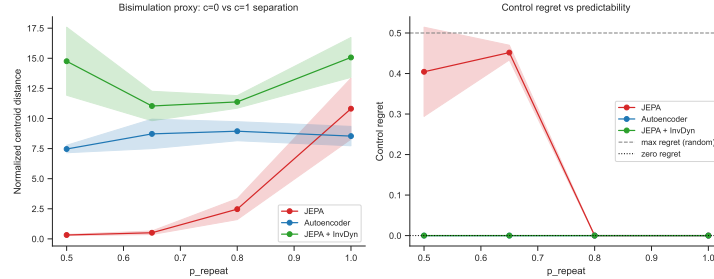


Figure 5: **Class separation** (centroid distance between $c = 0$ and $c = 1$ latent representations) for `jepa`, `recon`, and `oracle` at convergence (cell 4, $p_{\text{repeat}} = 0.5$, full training). The horizontal reference marks the analytical bisimulation distance (1.0). JEPA realizes only about 5% of the oracle’s class separation (0.105 versus 1.998); the resulting bisimulation error is 1.893.

- The `jepa_invdyn` result depends on the action policy: the cell-4 failure holds under random actions but need not hold under every informative policy.
- The regret-versus-information result (Figure 6) is de-circularized—both axes are independent of the linear probe—but was produced by a separate, reduced-budget run (2000 training steps on CPU, two seeds), not the full-training regime used elsewhere; and because the feature is one bit, the relationship is a sharp threshold, so the near-perfect Pearson correlation should be read as “retain $\Rightarrow$ zero regret, drop $\Rightarrow$ chance-level regret,” not as a smooth dose–response.

6.5 Future work

The natural next step is to demonstrate the failure mode on a pretrained V-JEPA 2-AC or DINO-WM encoder in a goal-not-given task. The minimal-reward-signal result (2% of transitions) suggests that even a weakly reward-conditioned adapter on a frozen JEPA encoder may be enough to prevent it.

A Environment implementation details

`QuadrantEnv` renders 12×12 grayscale frames: a 3×3 corner patch carries the feature bit, and the background is a sum of four spatial-frequency gratings whose phase advances deterministically each step, giving the encoder predictable structure to model. `FeatureSpec` records, per feature, whether it is controllable (toggled by the action) and whether it is relevant (sets the reward), and `make_cell(cell, predictability)` assembles the single-feature spec for each quadrant cell. Exogenous features evolve with $P(c_{t+1} = c_t) = p_{\text{repeat}}$; the optimal action is $a^* = c_t$ and the reward is 1 iff the action matches the relevant bit. The main experiments use 20000 training transitions, 4000 evaluation transitions, batch size 128, learning rate 10^{-3} , latent dimension 128, 4000 training steps, and seeds $\{0, 1, 2\}$. The two replication surface forms are `SwitchColorEnv` (32×32 , intensity patch over a sinusoidal grating) and `GridWorldHiddenRuleEnv` (bottom-right patch over a drifting texture with an agent marker). As a check on the predictability knob in `SwitchColorEnv`, JEPA recovers the feature at 1.00 ± 0.00 when $p_{\text{repeat}} = 1.0$ and only 0.61 ± 0.13 when $p_{\text{repeat}} = 0.5$, while reconstruction holds at 1.00 ± 0.00 throughout.

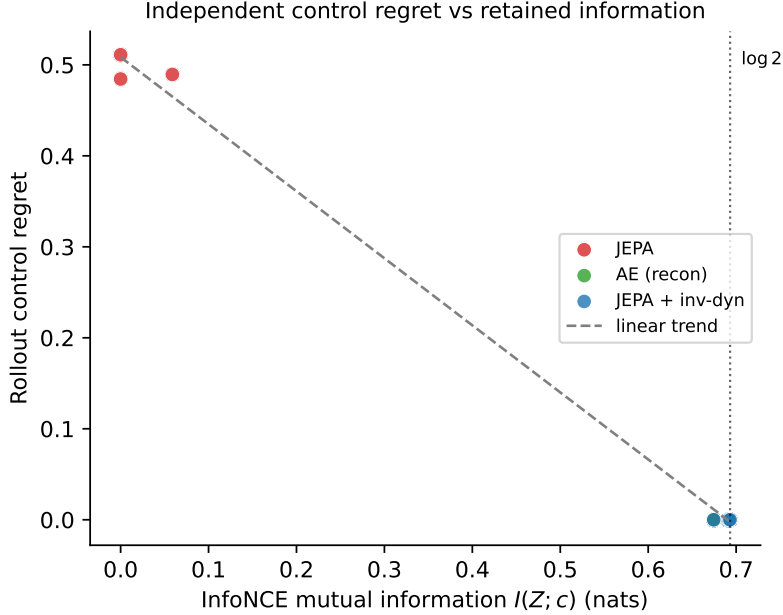


Figure 6: **Independent control regret versus retained information.** For each run of the SwitchColor predictability sweep (`jepa`, `ae`, `jepa_invdyn`; $p_{\text{repeat}} \in \{0.5, 0.65, 0.8, 1.0\}$; two seeds), the horizontal axis is the InfoNCE mutual information $I(Z; c)$ estimated by a separate critic, and the vertical axis is the regret of an *independent* behaviour-cloned policy rolled out in the environment. Neither quantity is derived from the linear probe, so the relationship is non-circular. Runs that retain the feature (MI saturating at $\log 2$) incur zero control regret; runs that drop it (MI ≈ 0) incur chance-level regret (≈ 0.49); the off-diagonal is empty (Pearson $r = -0.999$). This is a separate, reduced-budget run (2000 training steps), reported distinctly from the full-training numbers elsewhere in the paper (Section 6.4).

B InfoNCE estimator details

The mutual-information estimate is a separable InfoNCE critic (a small MLP) trained on a held-out split so the bound stays honest (van den Oord et al., 2018), run for 300 epochs. As a sanity check, the estimator returns approximately $\log 2 \approx 0.69$ nats when the feature is perfectly recoverable and approximately 0 nats when the feature is independent of the latent, matching the values observed for `recon/oracle` and for collapsed JEPa latents respectively (Table 2).

C Full results

Table 3 gives the full objective $\times$ cell matrix on `QuadrantEnv` (mean $\pm$ std over three seeds). In this matrix `jepa_invdyn` is trained on informative-action data (a^* with $\varepsilon = 0.2$ exploration); its cell-4 retention of 1.00 there tracks action informativeness, not reward grounding, and under random actions it drops to chance (Table 2). Table 4 gives the capacity sweep across both environments and the minimum reward-label fraction.

Table 3: Full objective $\times$ cell retention matrix. Values are linear probe accuracy (mean $\pm$ std over 3 seeds, chance = 0.50). Highlighted: cell 4 (exo + rel), the hard case where only reward-grounded signals retain the feature.

	Cell 1	Cell 2	Cell 3	Cell 4
Objective				
	ctrl+rel	ctrl+irr	exo+irr	exo+rel
recon	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00
jepa	0.51 $\pm$ 0.01	0.51 $\pm$ 0.01	0.52 $\pm$ 0.02	0.52 $\pm$ 0.02
jepa_ac	0.51 $\pm$ 0.01	0.51 $\pm$ 0.01	0.52 $\pm$ 0.03	0.52 $\pm$ 0.03
jepa_ctrl	0.67 $\pm$ 0.23	0.67 $\pm$ 0.23	0.52 $\pm$ 0.02	0.52 $\pm$ 0.02
jepa_invdyn	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	0.51 $\pm$ 0.03	1.00 $\pm$ 0.00
jepa_reward	1.00 $\pm$ 0.00	0.51 $\pm$ 0.01	0.52 $\pm$ 0.02	1.00 $\pm$ 0.00
oracle	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00

D Reproducibility

Results are deterministic given fixed seeds $\{0, 1, 2\}$ and were produced on a GPU (CUDA, PyTorch). An automated check (`verify.py`) asserts the encoder-identity invariant (matching `state_dict` keys and shapes across all objectives) and the evaluation-protocol invariant (disjoint evaluation seed; probes read the online encoder, never the EMA target), and exercises the InfoNCE sanity check above. Full reproduction instructions and the per-run result JSONs that back every full-training number in this paper are described in the accompanying `REPRODUCE.md`. The de-circularized regret-versus-information run (Figure 6) is reproduced separately by `regret_mi_experiment.py` (which records `mi_infonce` and `control_regret_rollout` per run to `regret_mi_sweep.json`) and `make_regret_mi_figure.py`; it uses a reduced budget of 2000 training steps on CPU and seeds $\{0, 1\}$.

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Table 4: (Top) Capacity sweep: cell-4 linear probe accuracy vs latent dimension for `jepa` and `jepa_reward` on both environments (3 seeds; values are means where per-seed data is available, else mean only). `jepa` stays near chance and `jepa_reward` stays near perfect across all sizes, ruling out capacity as the bottleneck. (Bottom) Minimum reward-signal fraction: the smallest `reward_label_fraction` for which `jepa_reward` retains cell 4 (probe acc ≥ 0.75) on Quadrant (3 seeds, mean shown).

latent_dim	Quadrant		SwitchColor	
	jepa	jepa_reward	jepa	jepa_reward
16	0.51	1.00	0.51	1.00
64	0.51	1.00	0.51	1.00
128	0.51	1.00	0.52	1.00
256	0.54	1.00	0.54	1.00
512	0.55	1.00	0.55	1.00
1024	0.56	0.91	0.56	0.93

Minimum reward-label fraction (Quadrant, cell 4)	
reward_frac	Cell-4 probe acc (mean)
0.01	0.72
0.02	0.77 ← min passing
0.05	1.00
0.1	1.00
0.25	1.00
0.5	1.00
1	1.00

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